



B.Tech. Degree VIII Semester Special Supplementary Examination July 2023

ME 19-205-0810 OPERATIONS RESEARCH (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Solve problems using optimization methods.
 CO2: Develop mathematical skills to approach real life, industrial problems.
 CO3: Inculcate ideas to solve problems on Transportation and Queueing models.
 CO4: Import and analyse case studies to solve future problems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

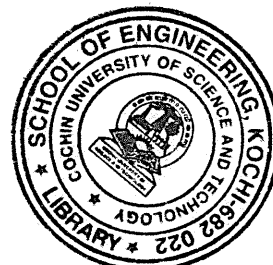
	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Define optimization problems. How do we determine optimal allocations in programming problems?	3		L1	1	1
(b) How will you understand the LPP has an unbounded solution while you solve it by graphical method?	3		L2	1	1
(c) What is degeneracy? Define cycling in simplex method.	3		L1	1	1
(d) How does Simplex terminate when the Linear Programming Problem is unbounded?	3		L2	1	1
(e) What are the objectives of Assignment problems?	3		L2	3	1
(f) Use the North West corner rule to obtain the initial basic feasible solution to a 3*4 matrix numerical (Transportation) problem of your choice.	3		L3	3	2
(g) Define Bellman's Principle of optimality.	3		L1	1	1
(h) What are the entities involved in a queueing system? Sketch and briefly explain.	3		L2	3	1

PART B

(4 × 12 = 48)

- II. (a) Define extreme point. Is it always necessary that the optimal solution of an LPP is an extreme point? Why? 5 L2 1 1
- (b) Bata Company makes two kinds of leather belts. A is a high-quality belt and B is of lower quality. The respective profits are ₹4 and ₹3 per belt. Each belt of type A requires twice as much time as belt of type B. If all belts were of type B, the Company could make 1000 per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 per day are available. There are only 700 buckles per day are available for belt B. Formulate the problem as a Linear Programming Problem. 7 L3 2 2

OR



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		Marks	BL	CO	PO
III.	(a) What is the indication in graphical method that an LPP has infeasible solutions?	3	L2	2	1
	(b) A Company buying scrap metal has two types of scrap available to them. The first type of scrap has 20% of metal A, 10% of impurity and 20% of metal B by weight. The second type of scrap has 30% of metal A, 10% of impurity, and 15% of metal B by weight. The company requires at least 120 Kg of metal A, at most 40 Kg of impurity and at least 90 Kg of metal B. The price for the two scraps are ₹200 and ₹300 per Kg respectively. Determine the optimum quantities of the two scraps to be purchased so that the requirements of the two metals and the restriction on impurity are satisfied at minimum cost. Use graphical method for solutions.	9	L3	3	2
IV.	(a) How will you understand the LPP has an unbounded solution while you solve it by Simplex method?	4	L2	1	1
	(b) Solve the following problem by Simplex method Maximize $z = 6x_1 + 8x_2$ Subjected to $x_1 + x_2 \leq 10$ $2x_1 + 3x_2 \leq 25$ $x_1 + 5x_2 \leq 35$ $x_1, x_2 \geq 0$	8	L3	1	2
OR					
V.	(a) What is duality in Linear Programming?	3	L2	1	1
	(b) Minimize $z = x_1 + x_2$ Subject to $2x_1 + x_2 \geq 4$ $x_1 + 7x_2 \geq 7$ $x_1, x_2 \geq 0$ Use two phase method for finding the solutions.	9	L3	1	2
VI.	Solve the given problem using VAM method and check for optimality condition using stepping stone or MODI method.	12	L4	3	2

	D1	D2	D3	D4	D5	supply
O1	12	4	9	5	9	55
O2	8	1	6	6	7	45
O3	1	12	4	7	7	30
O4	10	15	6	9	1	50
Demand	40	20	50	30	40	

OR

(Continued)

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- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| VII. (a) Summarise Hungarian Method of solving assignment problems with the help of a flow chart. | 6 | L2 | 3 | 2 |
| (b) Production department of a company has five employees with five jobs to be performed. The time each employee takes to perform job is given in the effectiveness matrix. How should the job be allocated so as to minimise the total man-hours? | 6 | L5 | 3 | 2 |

	I	II	II	IV	V
A	10	5	13	15	16
B	3	9	18	13	6
C	10	7	2	2	2
D	7	11	9	7	12
E	7	9	10	4	12

- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| VIII. A salesman wants to go from city 1 to city 10. The various possibilities and distances (in Km) are given in the following table. | 12 | L4 | 2 | 2 |

Arc	Distance	Arc	Distance	Arc	Distance	Arc	Distance
1-2	5	2-7	8	4-6	5	6-9	7
1-3	5	3-5	8	4-7	7	7-8	5
1-4	6	3-6	10	5-8	6	7-9	7
2-5	4	3-7	5	5-9	8	8-10	8
2-6	7	4-5	4	6-8	9	9-10	9

Use dynamic programming for solving the problem.

OR

- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| IX. (a) Define the terms Reneging, Jockeying, Balking and Collusion in queuing theory. | 4 | L1 | 4 | 1 |
| (b) Customers arrive at a doctors clinic at the rate of 8/hr (follows poisson arrival) and the doctor can serve at rate of 9/hr.(exponential service).
(i) What is the probability that a customer directly walks into the doctor's room?
(ii) What is the probability that there is no queue?
(iii) What is the expected number in the system?
(iv) What is the expected waiting time in the queue? | 8 | L4 | 4 | 2 |

Blooms's Taxonomy Levels

L1 – 10.8%, L2 – 27.5%, L3 – 30%, L4 – 26.7%, L5 – 5%

